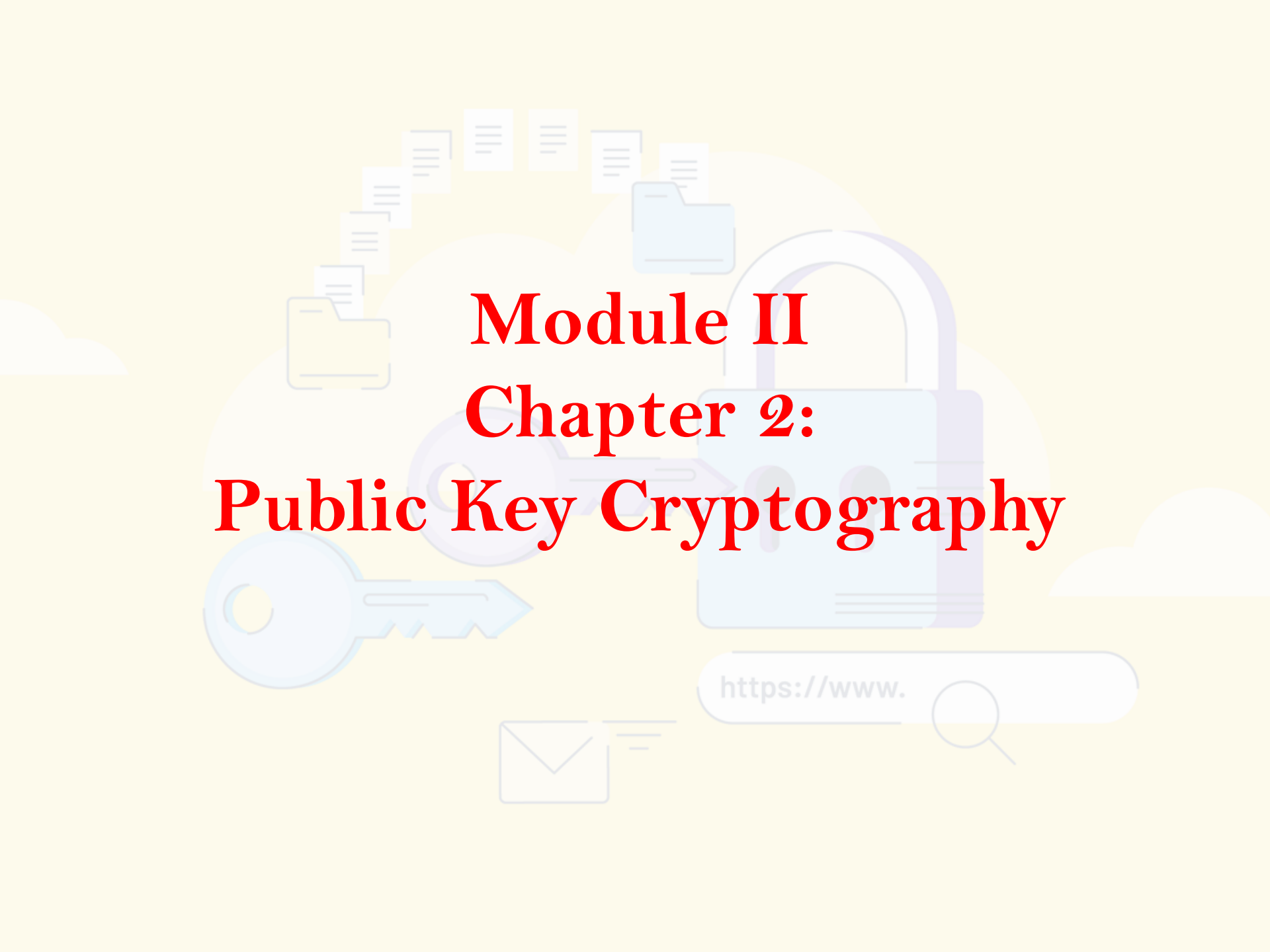


The background features a light yellow gradient with stylized white clouds. In the center, there is a large, faint blue padlock. To the left of the padlock is a large blue key. Above the padlock, a series of white document icons are arranged in a curved path. Below the padlock, there is a white rounded rectangle containing the text 'https://www.' followed by a magnifying glass icon. At the bottom center, there is a white envelope icon with two lines indicating motion.

Module II

Chapter 2:

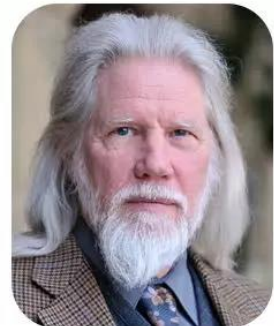
Public Key Cryptography

Introduction to Public key Cryptography:

- Public key cryptography also called as **asymmetric cryptography**.
- It was invented by **whitfield Diffie and Martin Hellman in 1976**. Sometimes this cryptography also called as **Diffie-Helman Encryption**.
- Public key algorithms are based on **mathematical problems** which admit no efficient solution that are inherent in certain integer factorization, discrete logarithm and Elliptic curve relations.

Public key Cryptosystem Principles:

- The concept of public key cryptography is invented for two most difficult problems of Symmetric key encryption.
 - **The Key Exchange Problem**
 - **The Trust Problem**



1. The Key Exchange Problem:

- The key exchange problem arises from the fact that communicating parties must somehow share **a secret key before any secure communication can be initiated**, and both parties must then **ensure** that the **key remains secret**. Of course, direct key exchange is not always feasible due to risk, inconvenience, and cost factors.

2. The Trust Problem:

- Ensuring the **integrity of received data and verifying the identity of the source of that data can be very important**. Means in the symmetric key cryptography system, receiver doesn't know whether the message is coming for particular sender.



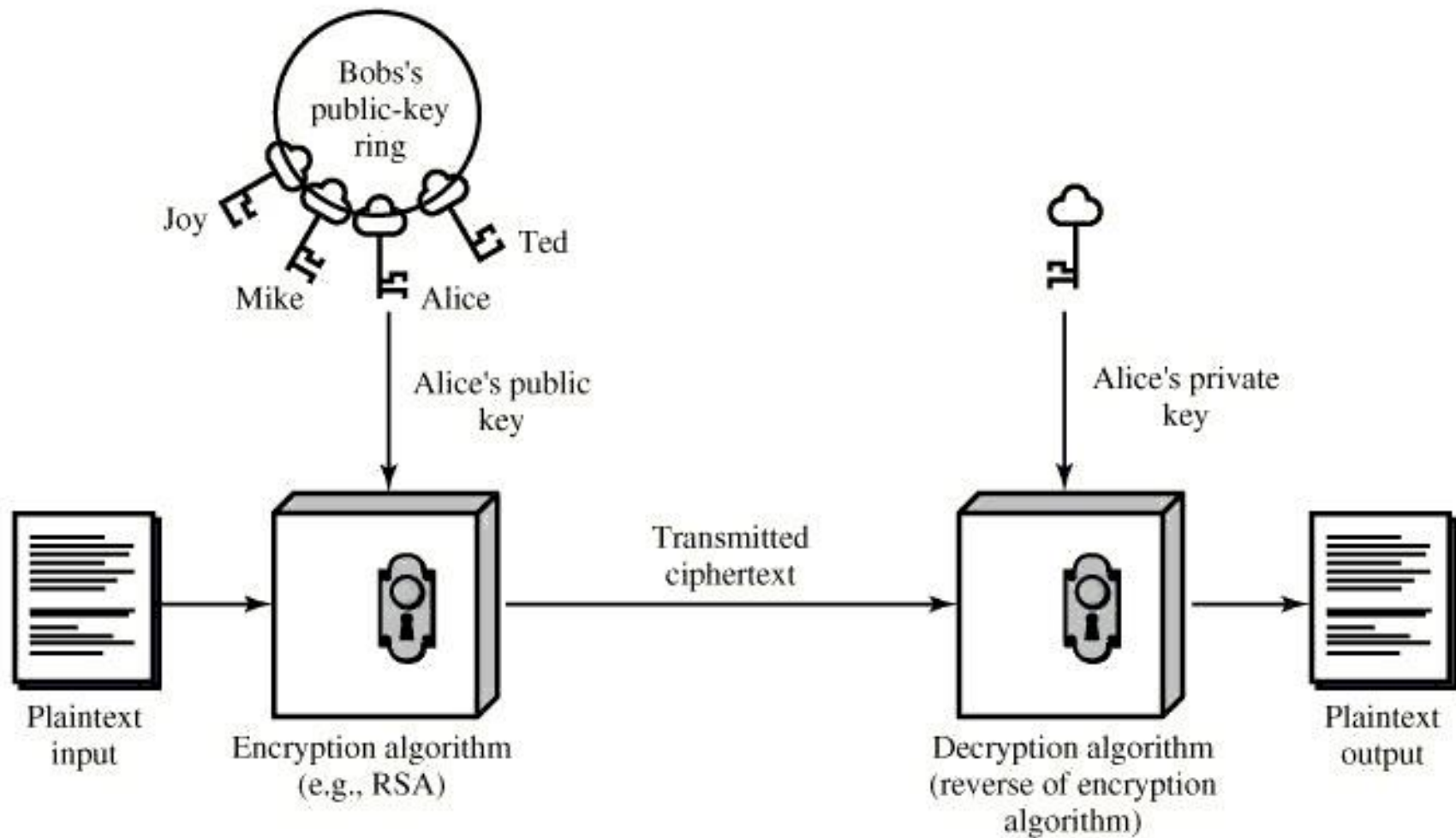
- This public key cryptosystem **uses two keys** as pair for encryption of plain text and Decryption of cipher text.
- These two keys are names as “**Public key**” and “**Private key**”.
- The **private key is kept secret** whereas **public key is distributed** widely.
- A message or text data which is encrypted with the public key can be decrypted only with the corresponding private-key.
- This two key system very useful in the areas of **confidentiality (secure) and authentication**



- A public-key encryption scheme has six ingredients

1	Plaintext	This is the readable message or data that is fed into the algorithm as input.
2	Encryption algorithm	The encryption algorithm performs various transformations on the plaintext.
3	Public key	This is a pair of keys that have been selected so that if one is used for encryption, the other is used for decryption. The exact transformations performed by the algorithm depend on the public or private key that is provided as input.
4	Private Key	
5	Ciphertext	This is the scrambled message produced as output. It depends on the plaintext and the key. For a given message, two different keys will produce two different ciphertexts.
6	Decryption algorithm	This algorithm accepts the ciphertext and the matching key and produces the original plaintext.

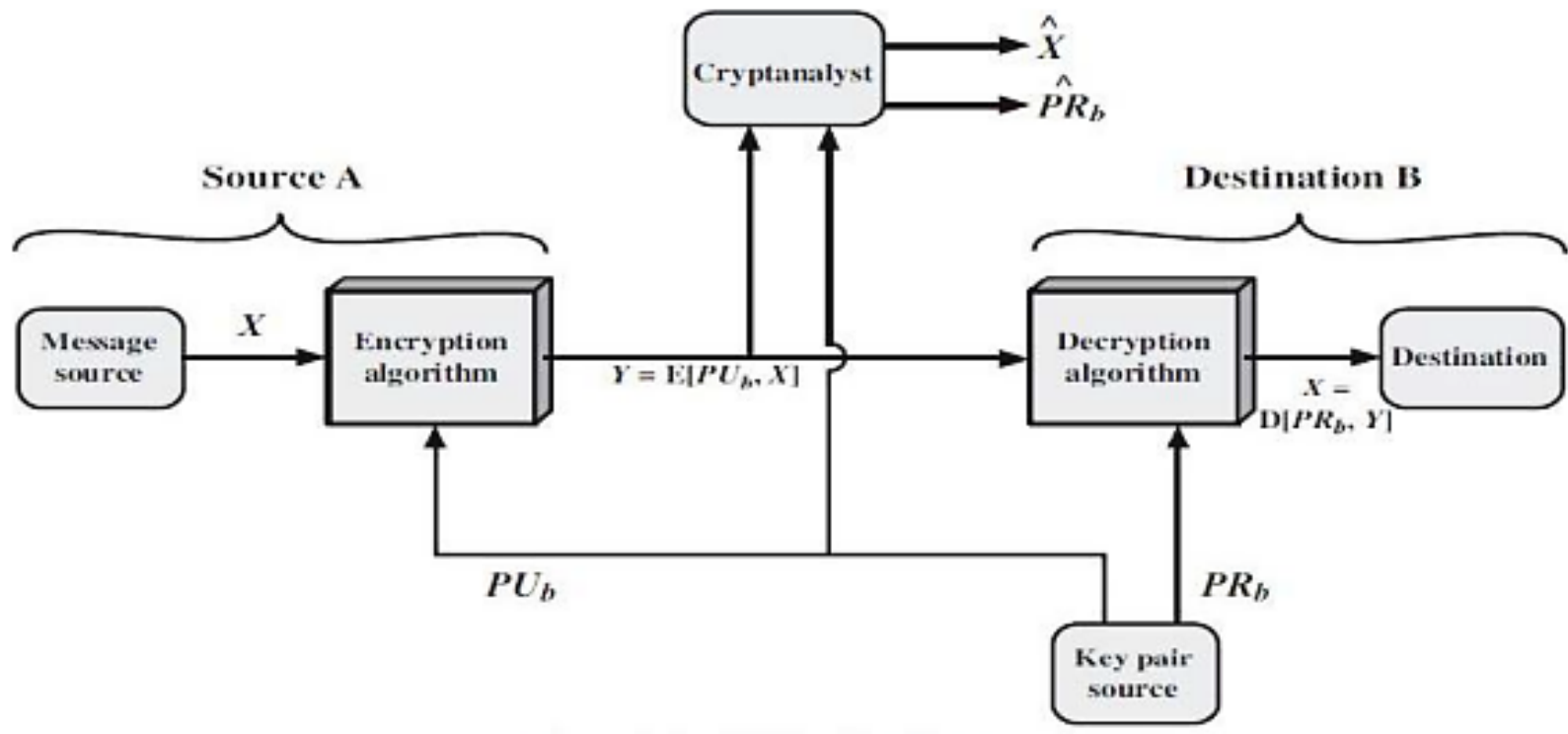
- Public key cryptography for providing confidentiality (secrecy)



(a) Encryption

The essential steps are the following.

1. Each user **generates a pair of keys** to be used for the encryption and decryption of messages.
2. Each user places one of the two keys in a public register or other accessible file. This is the public key. The companion key is kept private. So above fig states that each user maintains a collection of public keys obtained from others.
3. If Bob wishes to send a confidential message to Alice, **Bob encrypts the message using Alice's public key.**
4. When Alice receives the message, she decrypts it using her private key. No other recipient can decrypt the message because only Alice knows Alice's private key.



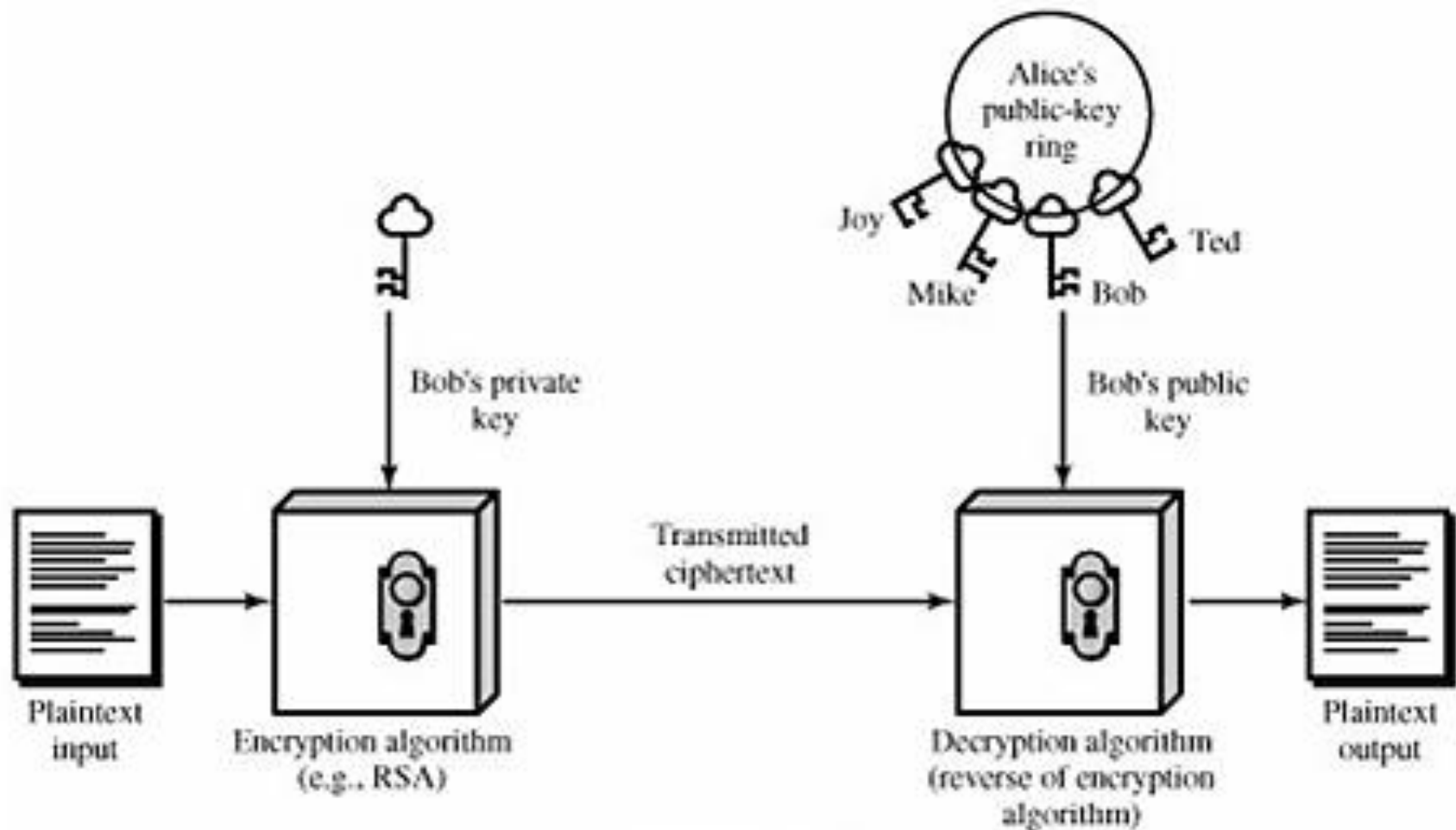
- There is some source A that produces a message in plaintext $X = [X_1, X_2, \dots, X_M]$.
- The M elements of X are letters in some finite alphabet.
- The message is intended for destination B. B generates a related pair of keys:
a public key, PU_b , and a private key, PR_b .
- **PR_b is known only to B, whereas PU_b is publicly available** and therefore accessible by A.
- With the message X and the encryption key PU_b as input, A forms the ciphertext $Y = [Y_1, Y_2, \dots, Y_N]$:

$$Y = E(PU_b, X)$$

- The intended receiver, in possession of the matching private key, is able to invert the transformation:

$$X = D(PR_b, Y)$$

- Public key cryptography for proving Authentication:



(b) Authentication

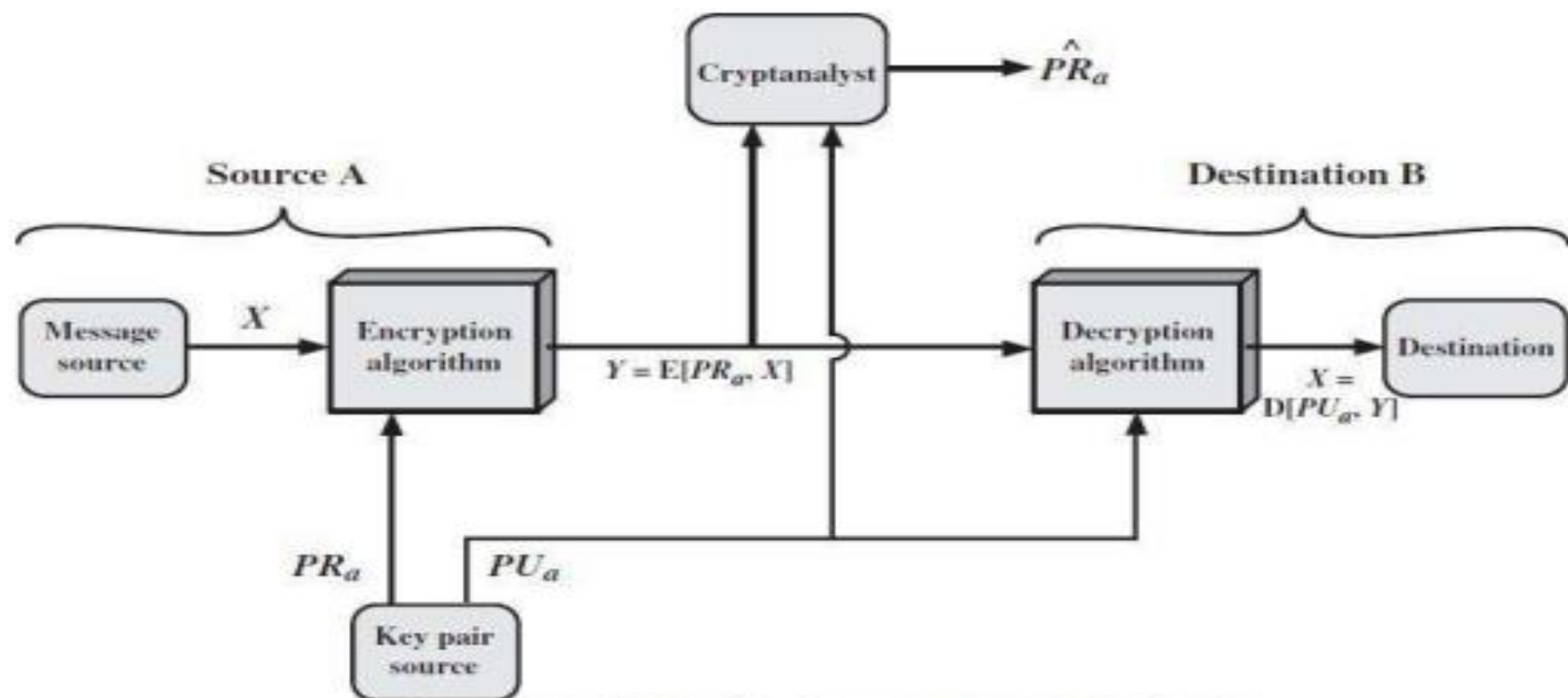


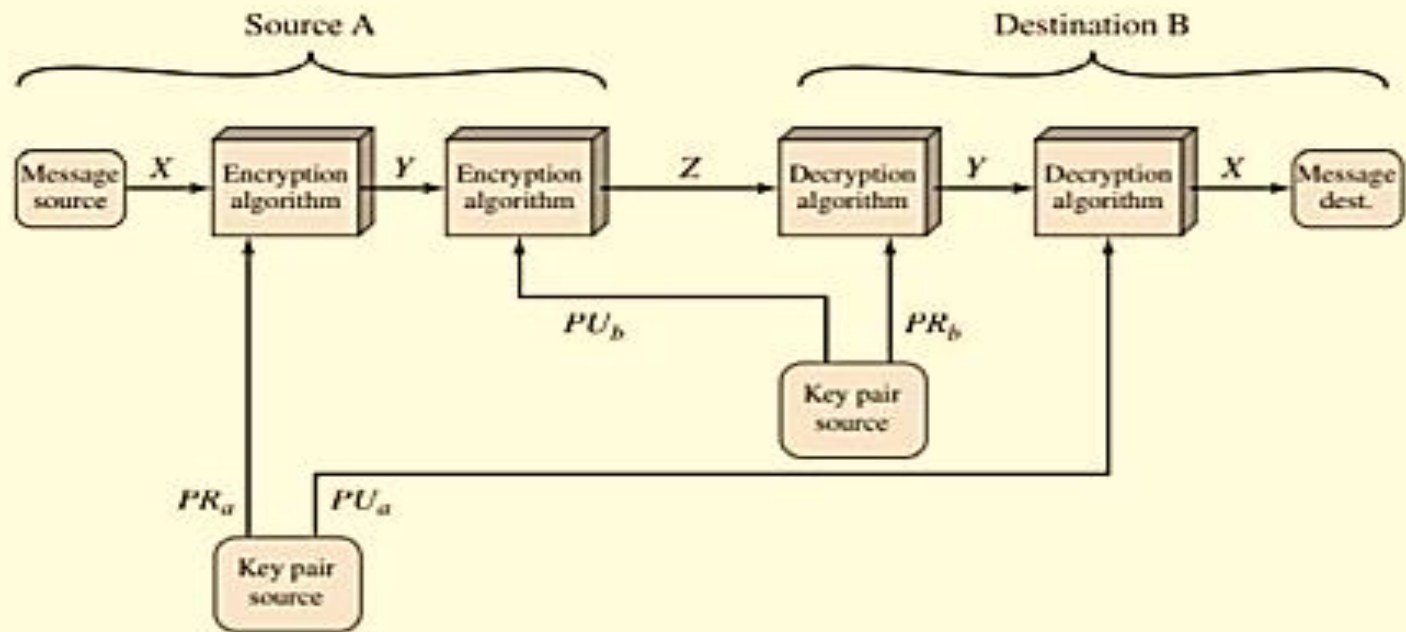
Figure 9.3 Public-Key Cryptosystem: Authentication

- The above diagrams show the use of public-key encryption to provide authentication:

$$Y = E(PR_a, X)$$

$$X = D(PU_a, Y)$$

- In this case, A prepares a message to B and encrypts it using A's private key before transmitting it. B can decrypt the message using A's public key.
- Because the message was encrypted using A's private key, only A could have prepared the message. Therefore, the entire encrypted message serves as a **digital signature**.
- It is impossible to alter the message without access to A's private key, so the message is **authenticated both in terms of source and in terms of data integrity**.



- It is, however, possible to provide both the authentication function and confidentiality by a double use of the public-key scheme (above figure):

$$Z = E(PU_b, E(PR_a, X))$$

$$X = D(PU_a, D(PR_b, Z))$$

- In this case, we begin as before by encrypting a message, using the **sender's private key. This provides the digital signature.**
- Next, we **encrypt again, using the receiver's public key.** The final ciphertext can be decrypted only by the intended receiver, who alone has the matching private key. Thus, **confidentiality is provided.**

Applications for Public-Key Cryptosystems:

- **Encryption /decryption:** The sender encrypts a message with the recipient's public key.
- **Digital signature:** The sender “signs” a message with its private key. Signing is achieved by a cryptographic algorithm applied to the message or to a small block of data that is a function of the message.
- **Key exchange:** Two sides cooperate to exchange a session key. Several different approaches are possible, involving the private key(s) of one or both parties.

Algorithm	Encryption/Decryption	Digital Signature	Key Exchange
RSA	Yes	Yes	Yes
Elliptic Curve	Yes	Yes	Yes
Diffie-Hellman	No	No	Yes
DSS	No	Yes	No

Advantages

- **Authentication Assurance**

Public key cryptography strengthens authentication by enabling the verification of message authenticity through digital signatures. This ensures that recipients can confidently validate the origin of messages, enhancing the overall security of communication channels.

- **Simplified Key Distribution**

Public key cryptography addresses the complexities of key distribution by encouraging users to openly share their public keys. This eliminates the need for secure pre-shared key channels, offering a more convenient and scalable approach to managing encryption keys.

- **Non-Repudiation via Digital Signatures**

Public key cryptography introduces non-repudiation features comparable to physical signatures. Messages digitally signed serve as unequivocal acknowledgments, preventing senders from disowning their communications. This strengthens the accountability and trustworthiness of digital interactions.

Disadvantages

- **Security Risks of Private Key Exposure**

If an attacker gains access to your private key, there is a significant security risk as they can decrypt and read all your messages. The confidentiality of your communication becomes compromised, emphasizing the importance of safeguarding private keys.

- **Impact of Private Key Loss**

The loss of a private key poses a critical challenge, as it renders you unable to decrypt received messages. This loss of access can result in the permanent inability to retrieve or understand sensitive information, underscoring the need for secure key management practices.

- **Performance Considerations**

Public key cryptography tends to operate at a **slower pace compared to symmetric cryptography**. This method may **not be suitable for decrypting bulk messages efficiently**, highlighting a potential drawback in terms of speed and performance, particularly in scenarios where large volumes of data need decryption.

RSA Algorithm

- The **RSA algorithm** is a public-key cryptography technique used to secure data transmission.
- It was introduced in 1977 by **Ron Rivest**, **Adi Shamir**, and **Leonard Adleman** at the Massachusetts Institute of Technology.
- RSA is widely used in secure communication protocols such as SSL and TLS.



RSA Algorithm

- RSA is an **asymmetric encryption algorithm**, meaning it uses **two keys**:
 -  **Public Key (for encryption)**
 -  **Private Key (for decryption)**
- Public key is shared with everyone. Private key is kept secret.

RSA Algorithm – Step by Step

Step 1: Choose Two Prime Numbers

- Select two large prime numbers: ***p and q***
- Example:
 $p = 3$
 $q = 11$
- (Real systems use very large primes like 1024-bit or 2048-bit numbers.)

Step 2: Calculate n

- $n = p \times q$
- Example:
 - $n = 3 \times 11 = 33$
 - This **n** is used in both public and private keys.

Step 3: Compute Euler's Totient Function

- $\phi(n) = (p-1)(q-1)$
- Example:
 - $\phi(n) = (3-1)(11-1) = 2 \times 10 = 20$

Step 4: Choose Public Key (e)

- Choose a number **e** such that: $1 < e < \phi(n)$
- e and $\phi(n)$ are **co-prime (GCD = 1)**
- Example:
 $e = 3$
 $\text{GCD}(3, 20) = 1$
- Public Key = (e, n) -> Public Key = (3, 33)

Step 5: Find Private Key (d)

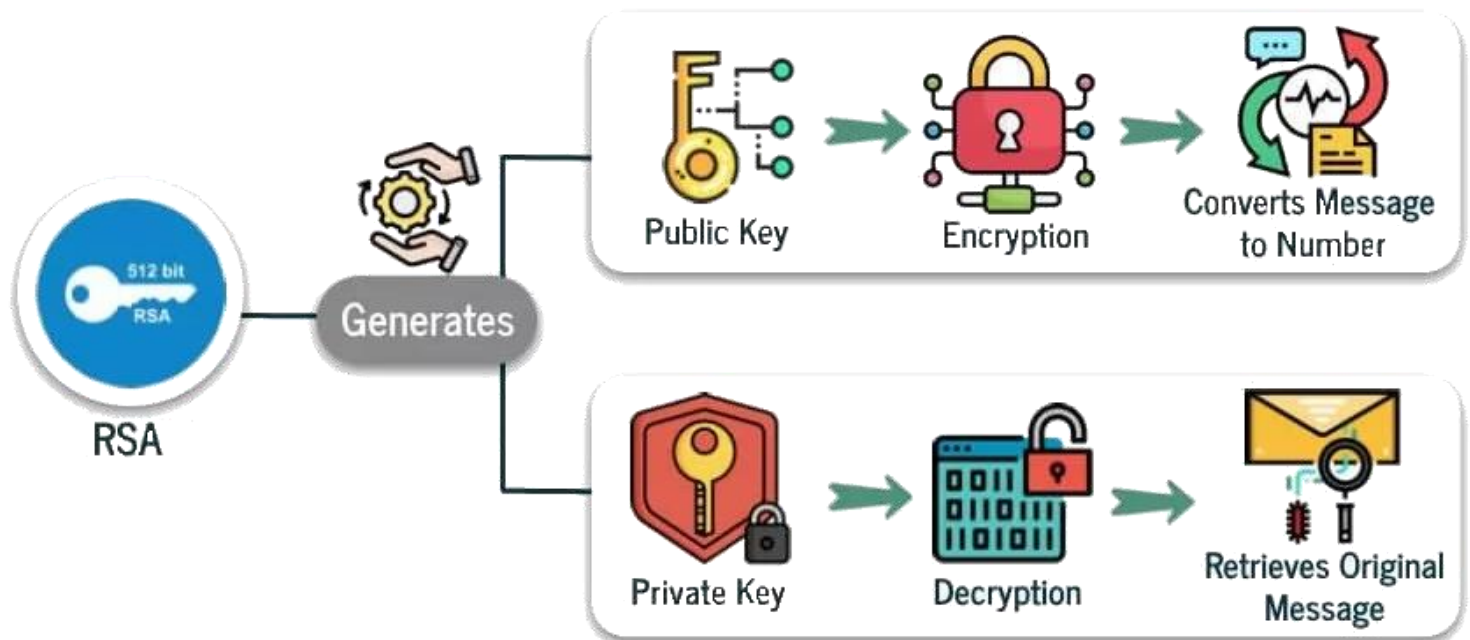
- Find **d** such that: $d \times e \equiv 1 \pmod{\phi(n)}$
- This means: $d \times 3 \equiv 1 \pmod{20}$
- Solve:
- $3 \times 7 = 21$
- $21 \bmod 20 = 1$
- So, $d = 7$
- Private Key = (7, 33)

Encryption Process

- Suppose message = 4
- Encryption formula:
 - $C = M^e \bmod n$
- $C = 4^3 \bmod 33$
- $C = 64 \bmod 33 = 31$
- Encrypted message = 31

Decryption Process

- Decryption formula:
 - $M = C^d \bmod n$
- $M = 31^7 \bmod 33$
- $M = 317 \bmod 33$
- Result = 4
- Original message recovered!



RSA key exchange

ALICE

Chooses two prime numbers (p and q)

NOTE: $p \neq q$

Calculates modulus (n)
 $n = p \times q$

Calculates totient of mod n ($\phi(n)$)
 $\phi(n) = (p-1) \times (q-1)$

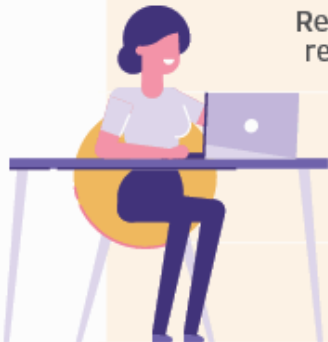
Chooses public key (e)

NOTE: e MUST BE LESS THAN AND COPRIME WITH $\phi(n)$

Calculates private key (d) using
Extended Euclidean algorithm

NOTE: d IS THE MULTIPLICATIVE INVERSE OF $e \bmod \phi(n)$

Releases (e, n) as public key,
retains (d, n) as private key



Alice decrypts M using her private key (d, n)
 $M = C^d \bmod n$

BOB



Bob wants to send Alice a message (M)

Bob obtains Alice's public key (e, n)

Bob encrypts M into ciphertext (C):
 $C = M^e \bmod n$

Steps of RSA algorithm:

Step 1 : Select 2 prime numbers p & q

Step 2 : Calculate $n=pq$

Step 3 : Calculate $\phi(n)=(p-1)(q-1)$

Step 4 : Select or find integer e (public key) which is relatively prime to $\phi(n)$. i.e., e with $\gcd(\phi(n), e)=1$ where $1 < e < \phi(n)$.

Step 5 : Calculate “ d ” (private key) by using following condition.

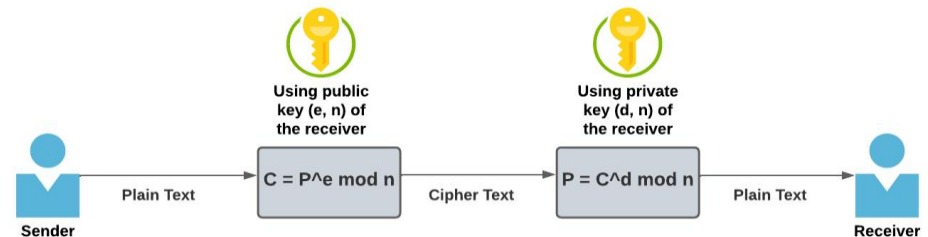
$$d < \phi(n). \text{ i.e; } \mathbf{ed \equiv 1 \pmod{\phi(n)}}$$

Step 6 : Perform encryption by using.

$$\mathbf{C = M^e \pmod n}$$

Step 7: Perform Decryption by using

$$\mathbf{M = C^d \pmod n}$$



Example:

- Let us solve the RSA example clearly and step-by-step using:
- $p=17$
- $q=11$
- Message $M=88$

Step 1: Compute n

- $n=p \times q$
- $n=17 \times 11=187$

Step 2: Compute Euler's Totient Function

- $\phi(n)=(p-1)(q-1)$
- $\phi(n)=(17-1)(11-1)$
- $\phi(n)=16 \times 10=160$

Step 3: Choose Public Key e

- We must choose e such that: $1 < e < 160$
- $\text{GCD}(e, 160) = 1$
- Let us choose: $e = 7$
- Since $\text{GCD}(7, 160) = 1$, So it is valid.
- **Public Key:** $(e, n) = (7, 187)$

Step 4: Find Private Key d

- We must find d such that: $d \times e \equiv 1 \pmod{160}$
- So, $d \times 7 \equiv 1 \pmod{160}$
- Using modular inverse: $7 \times 23 = 161$
- $161 \pmod{160} = 1$
- So, $d = 23$
- **Private Key:**
- $(d, n) = (23, 187)$

Step 5: Encryption

Formula:

$$C = M^e \mod n$$

$$C = 88^7 \mod 187$$

Using modular reduction:

$$88^2 \mod 187 = 7744 \mod 187 = 77$$

$$88^4 \mod 187 = 77^2 = 5929 \mod 187 = 132$$

$$88^7 = 88^4 \times 88^2 \times 88$$

$$= 132 \times 77 \times 88 \mod 187$$

After simplifying stepwise:

$$C = 11$$

 Encrypted Message:

$$C = 11$$

Step 6: Decryption

Formula:

$$M = C^d \mod n$$

$$M = 11^{23} \mod 187$$

After modular exponentiation calculation:

$$M = 88$$

 Original message recovered successfully.

Advantages of RSA

- **No Key Sharing:** RSA encryption depends on using the receiver's public key, so you don't have to share any secret key to receive messages from others.
- **Proof of Authenticity:** Since the key pairs are related to each other, a receiver can't intercept the message since they won't have the correct private key to decrypt the information.
- **Faster Encryption:** The encryption process is faster than that of the DSA algorithm.
- **Data Can't Be Modified:** Data will be tamper-proof in transit since meddling with the data will alter the usage of the keys. And the private key won't be able to decrypt the information, hence alerting the receiver of manipulation.



No need of sharing
secret keys



Proof of owner's
authenticity



Faster Encryption
than DSA



Data can't be
modified in transit

RSA applications and use cases

- Secure communications.
- Digital certificates.
- TLS support.
- Secure key exchange.
- Financial transactions.
- Software licensing.

Security vulnerabilities and RSA attacks

- Key generation issues.
- Compromised keys.
- Side-channel attacks.
- Fault-based attacks.

How to avoid RSA vulnerabilities

- Strong key sizes.
- Strong prime number generation.
- Tamper-resistant hardware.
- Padding schemes.
- Vulnerability monitoring.
- Alternative algorithms.

Key Management in Cryptography

- In cryptography, it is a very monotonous task to distribute the public and private keys between sender and receiver.
- If the key is known to the third party (forger/eavesdropper) then the whole security mechanism becomes worthless. So, there comes the need to secure the exchange of keys.

What is Key Management?

- Key management refers to the **processes and procedures** involved in **generating, storing, distributing, and managing cryptographic keys** used in cryptographic algorithms to protect sensitive data.
- It **ensures** that keys used to **protect sensitive data** are kept safe from **unauthorized access or loss**.
- Good key management helps **maintain the security** of encrypted information and is important for protecting **digital assets from cyber threats**.
- Effective key management is crucial for ensuring the confidentiality, integrity, and availability of encrypted information by securing cryptographic keys from unauthorized access, loss, or compromise.

How Cryptographic Keys Works?

- Cryptographic keys are special codes that protect information by **locking (encrypting) and unlocking (decrypting)** it.
- In **symmetric key cryptography**, a single shared key does both jobs, so the same key must be kept secret between users.
- In **asymmetric key cryptography**, there are two keys: a public key that anyone can use to encrypt messages or verify signatures, and a private key that only the owner uses to decrypt messages or create signatures.
- This makes it easier to share the public key openly while keeping the private key secret.
- These keys are crucial for secure communication, like when you visit a secure website (HTTPS), where they help encrypt your data and keep it safe from eavesdroppers and criminals.
- So, to manage these keys properly is vital to keep digital information secure and dependable.

Types of Key Management

There are two aspects of Key Management:

- Distribution of public keys.
- Use of public-key encryption to distribute secrets.

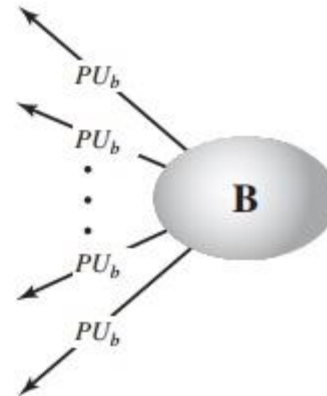
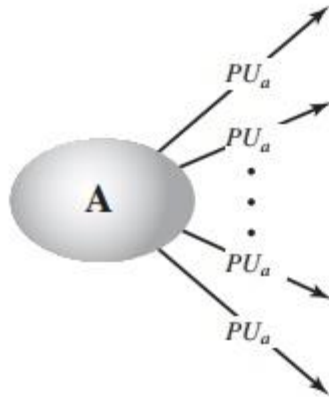
1. Distribution of Public Key

The public key can be distributed in four ways:

- **Public announcement**
- **Publicly available directory**
- **Public-key authority**
- **Public-key certificates.**

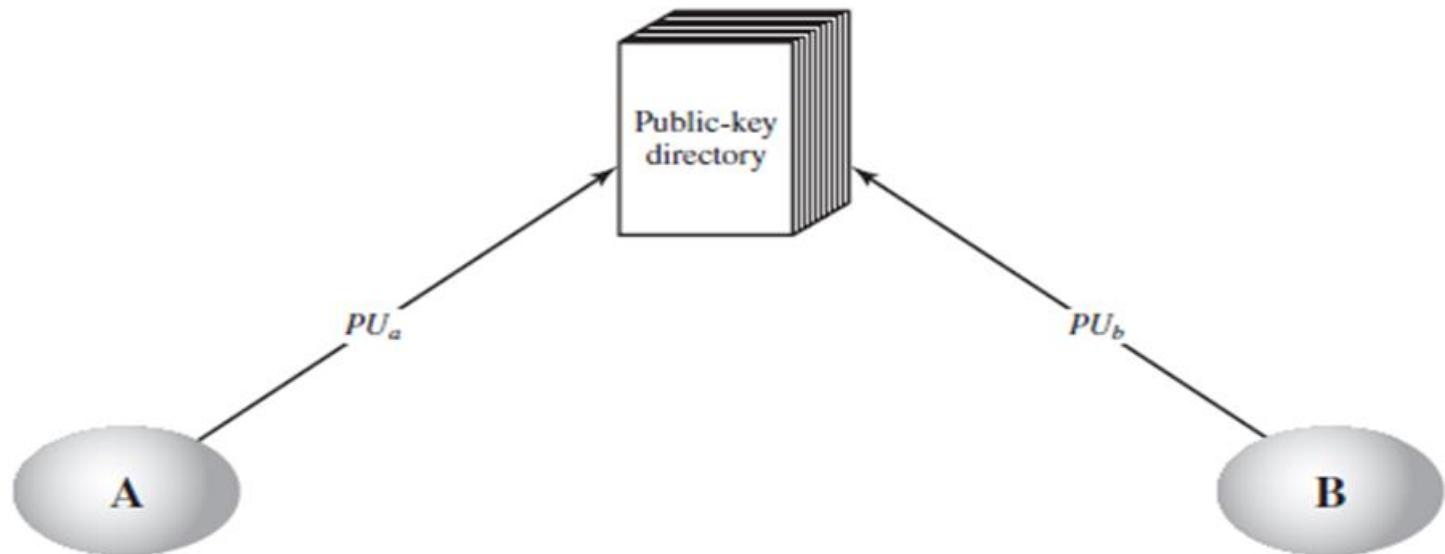
1. Public Announcement:

- Here the public key is **broadcast to everyone**.
- The **major weakness** of this method is a **forgery**.
- **Anyone** can create a key claiming to be someone else and **broadcast it**.
- Until forgery is discovered can masquerade as claimed user.



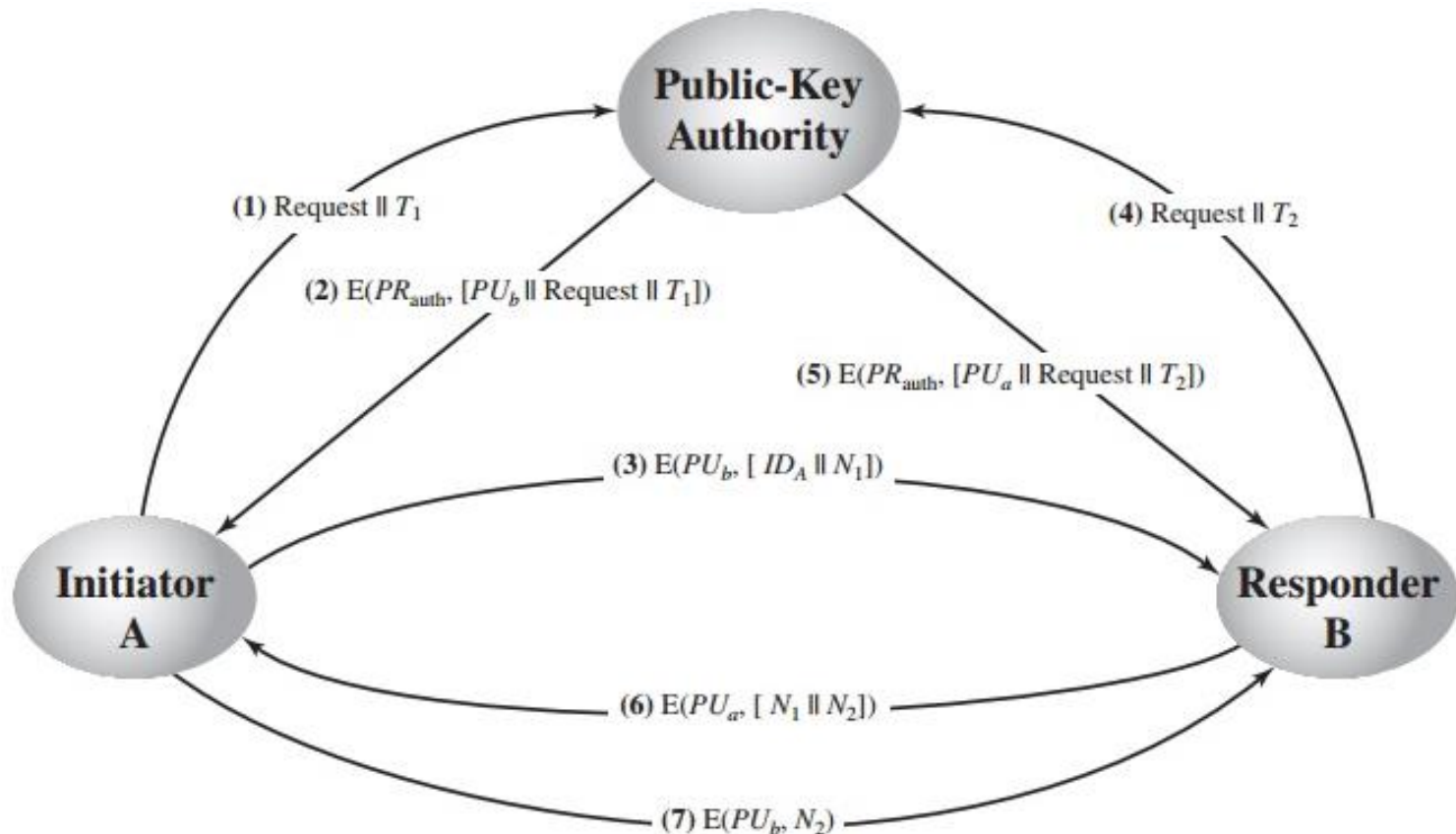
2. Publicly Available Directory:

- In this type, the **public key** is stored in a **public directory**.
- Directories are trusted here, with properties like **Participant Registration, access and allow to modify values at any time, contains entries like {name, public-key}**.
- Directories can be **accessed electronically** still vulnerable to forgery or tampering.



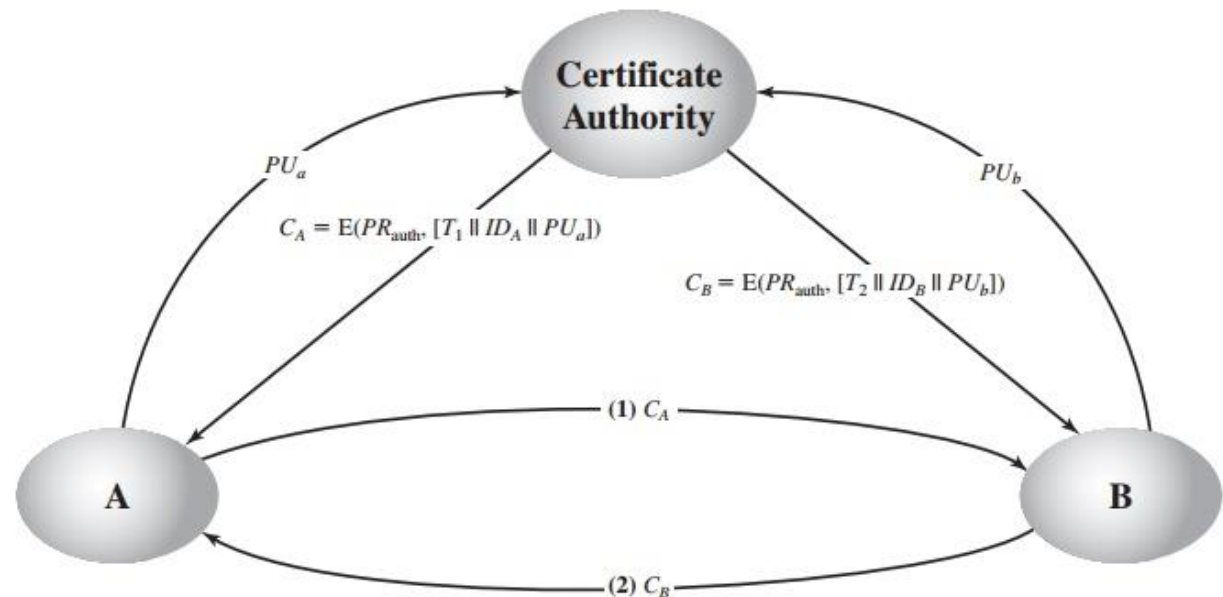
3. Public Key Authority:

- It is similar to the directory but, improves security by **tightening control over the distribution of keys from the directory.**
- It requires users to know the **public key for the directory.**
- Whenever the keys are needed, real-time access to the directory is made by the user to obtain any desired public key securely.

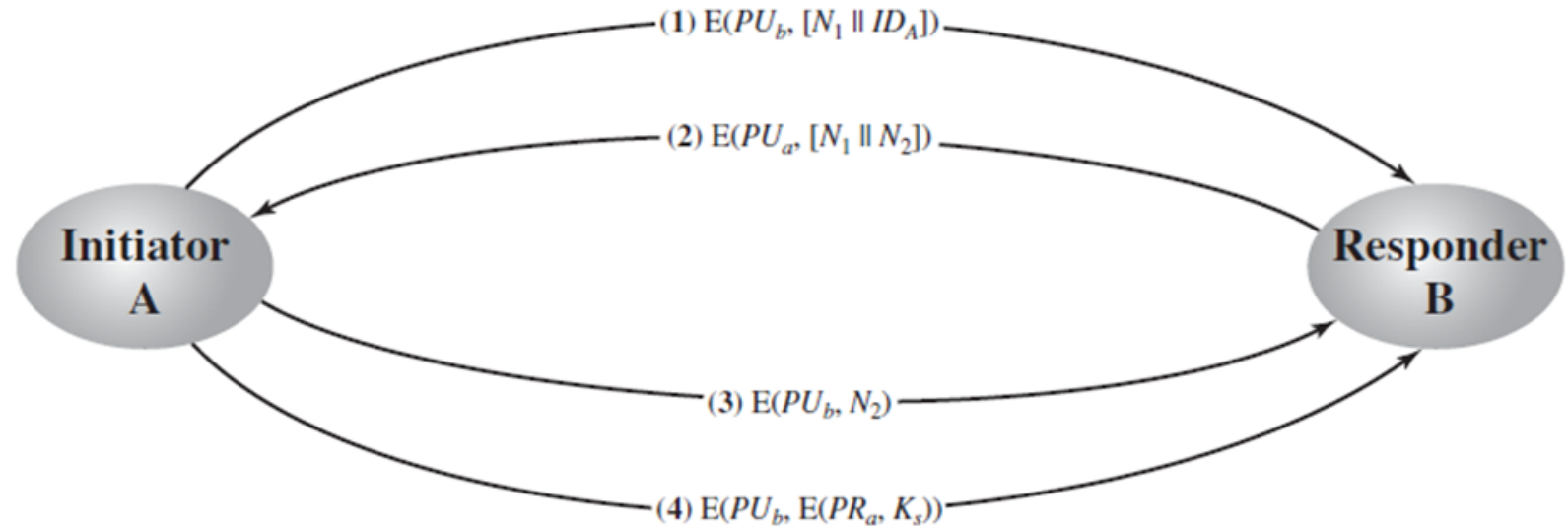


4. Public Certification:

- This time authority **provides a certificate** (which binds an identity to the public key) **to allow key exchange** without real-time access to the public authority each time.
- The certificate is accompanied by some other info such as **period of validity, rights of use, etc.**
- All of this content is signed by the private key of the certificate authority and it can be verified by anyone possessing the authority's public key.
- First sender and receiver both request CA for a certificate which contains a public key and other information and then they can exchange these certificates and can start communication.



2. Secret Key Distribution with Confidentiality and Authentication



- i. A uses B's public key to encrypt a message to B containing an identifier of A (ID_A) and a nonce (N_1), which is used to identify this transaction uniquely.
- ii. B sends a message to A encrypted with PU_a and containing A's nonce (N_1) as well as a new nonce generated by B (N_2)
- iii. A returns N_2 encrypted using B's public key, to assure B that its correspondent is A. A selects a secret key K_s and sends $M = E(PU_b, E(PR_a, K_s))$ to B. Encryption of this message with B's public key ensures that only B can read it; encryption with A's private key ensures that only A could have sent it.
- iv. B computes $D(PU_a, D(PR_b, M))$ to recover the secret key.

Key Management Lifecycle

- The key management lifecycle outlines the stages through which cryptographic keys are generated, used, and eventually retired or destroyed.
- Proper management of these keys is critical to ensuring the security of cryptographic systems.
- Here's an overview of each stage:

1. Key Generation:

- **Creation:** Keys are created using secure algorithms to ensure randomness and strength.
- **Initialization:** Keys are initialized with specific parameters required for their intended use (e.g., length, algorithm).

2. Key Distribution:

- **Sharing:** For symmetric keys, secure methods must be used to share the key between parties.
- **Publication:** For asymmetric keys, the public key is shared openly, while the private key remains confidential.

3. Key Storage:

- **Protection:** Keys must be stored securely, typically in hardware security modules (HSMs) or encrypted key stores, to prevent unauthorized access.
- **Access Control:** Only authorized users or systems should be able to access keys.

4. Key Usage:

- **Application:** Keys are used for their intended cryptographic functions, such as encrypting/decrypting data or signing/verifying messages.
- **Monitoring:** Usage is monitored to detect any unusual or unauthorized activities.

5. Key Rotation:

- **Updating:** Keys are periodically updated to reduce the risk of exposure or compromise.
- **Re-Keying:** New keys are generated and distributed, replacing old ones while ensuring continuity of service.

6. Key Revocation:

- **Invalidation:** Keys that are no longer secure or needed are invalidated.
- **Revocation Notices:** For public keys, revocation certificates or notices are distributed to inform others that the key should no longer be trusted.

7. Key Archival:

- **Storage:** Old keys are securely archived for future reference or compliance purposes.
- **Access Restrictions:** Archived keys are kept in a secure location with restricted access.

8. Key Destruction:

- **Erasure:** When keys are no longer needed, they are securely destroyed to prevent any possibility of recovery.
- **Verification:** The destruction process is verified to ensure that no copies remain.

DIFFIE- HELLMAN KEY EXCHANGE

- The Diffie–Hellman key exchange is a foundational public-key cryptographic protocol that **enables two parties to establish a shared secret over an insecure communication channel**.
- It was introduced by Whitfield Diffie and Martin Hellman in 1976 and remains a conceptual cornerstone of modern cryptography.
- Unlike encryption algorithms that directly secure messages, Diffie–Hellman is specifically designed for secure key establishment.

Conceptual Overview

The protocol allows two users to:

- Publicly agree on two numbers.
- Privately choose their own secret numbers.
- Exchange computed values.
- Independently compute the same shared secret.
- Even if an **attacker** intercepts the exchanged values, **deriving the secret key is computationally infeasible** due to the hardness of the discrete logarithm problem.

Method of Operation in Diffie-Hellman Key Exchange

Step 1: Public Parameters

Two users (Alice and Bob) agree publicly on:

- Prime number: $p=23$
- Base (a primitive root modulo p): $g=5$
- These values are not secret.

Step 2: Private Keys

- Alice chooses a secret number: $a=6$
- Bob chooses a secret number: $b=15$
- These numbers are kept private.

Step 3: Compute Public Values

Alice computes:

$$A = g^a \mod p$$

$$A = 5^6 \mod 23 = 15625 \mod 23 = 8$$

Bob computes:

$$B = g^b \mod p$$

$$B = 5^{15} \mod 23 = 19$$

They exchange:

- Alice sends **8** to Bob.
- Bob sends **19** to Alice.

Step 4: Compute Shared Secret

Alice computes:

$$S = B^a \mod p$$

$$S = 19^6 \mod 23 = 2$$

Bob computes:

$$S = A^b \mod p$$

$$S = 8^{15} \mod 23 = 2$$

Both obtain the same secret:

$$\mathbf{Shared\ Key = 2}$$

This shared value can now be used as a symmetric encryption key.

Why It Works

- Mathematically:

$$(g^a)^b \mod p = (g^b)^a \mod p = g^{ab} \mod p$$

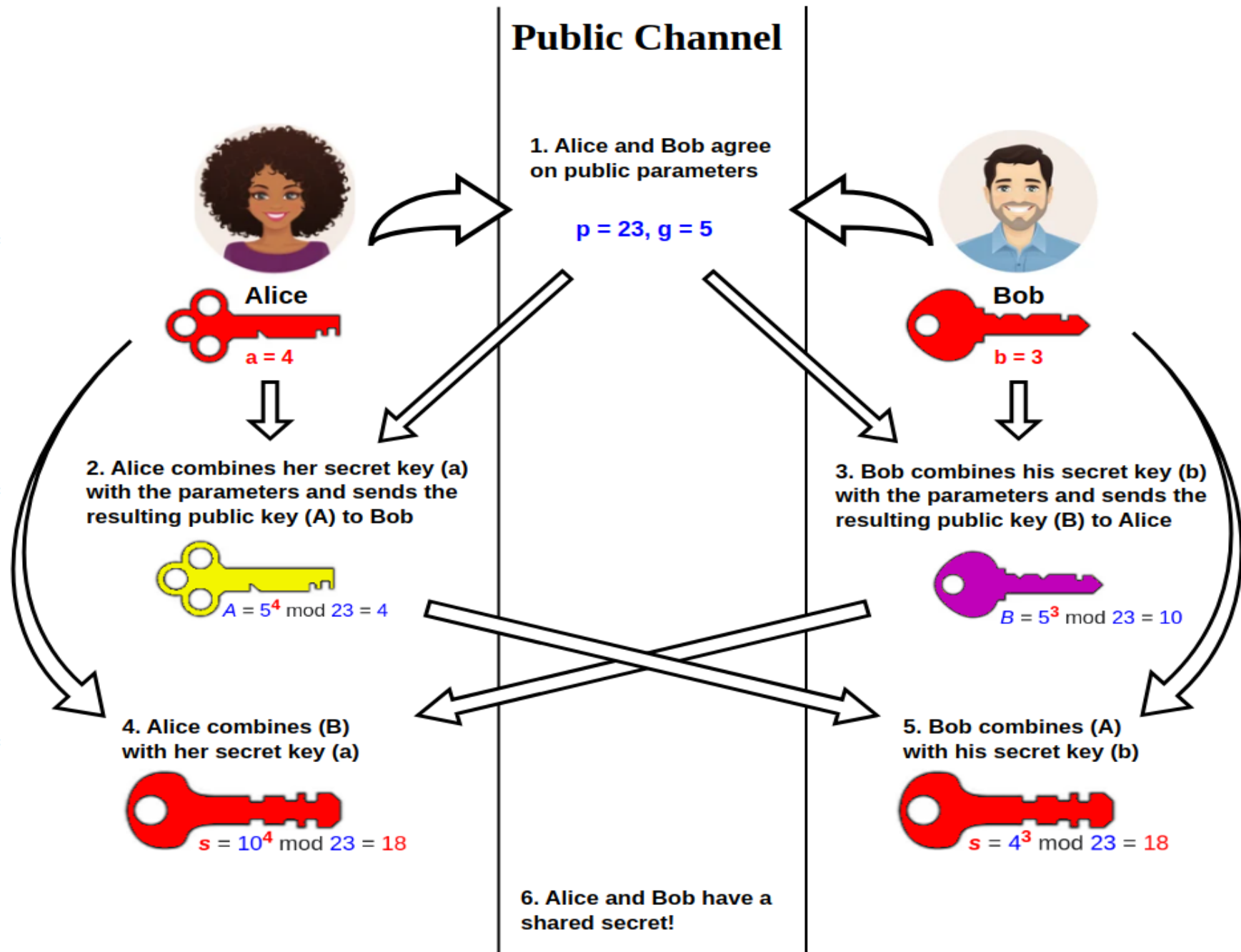
- Thus, both sides independently compute the same value without revealing their private numbers.

Security Basis

The security of Diffie–Hellman depends on:

- The **Discrete Logarithm Problem**
- Large prime modulus (typically 2048 bits or more)
- An attacker who sees:
 - p
 - g
 - A
 - B
 - cannot feasibly compute a or b when proper key sizes are used.

Example 2:



Example 3:

Public Parameters (Known to Everyone)

- Prime number: $p=11$
- Primitive root of 11: $g=2$

(2 is a primitive root of 11 because its powers generate all numbers from 1 to 10.)

Step 2: Private Keys (Secret)

- Alice chooses: $a=3$
- Bob chooses: $b=6$
- These values are never shared.

Step 3: Compute Public Values

Alice Computes:

$$A = g^a \mod p$$

$$A = 2^3 \mod 11$$

$$2^3 = 8$$

So, Alice sends $A = 8$ to Bob.

Bob Computes:

$$B = g^b \mod p$$

$$B = 2^6 \mod 11$$

$$2^6 = 64$$

$$64 \mod 11 = 9$$

So, Bob sends $B = 9$ to Alice.

Step 4: Compute Shared Secret

Alice Computes:

$$S = B^a \mod p$$

$$S = 9^3 \mod 11$$

$$9^2 = 81 \equiv 4 \pmod{11}$$

$$9^3 = 4 \times 9 = 36 \equiv 3 \pmod{11}$$

So, Alice gets $S = 3$.

Bob Computes:

$$S = A^b \pmod{p}$$

$$S = 8^6 \pmod{11}$$

$$8^2 = 64 \equiv 9 \pmod{11}$$

$$8^4 = 9^2 = 81 \equiv 4 \pmod{11}$$

$$8^6 = 8^4 \times 8^2 = 4 \times 9 = 36 \equiv 3 \pmod{11}$$

So, Bob also gets $S = 3$.

Final Shared Key

Shared Secret Key=3

Both parties independently computed the same value without ever transmitting their private numbers.

Diffie-Hellman key exchange

ALICE

BOB

Alice and Bob agree on two positive integers (p and q)

NOTE: p MUST BE A PRIME NUMBER, AND q MUST BE A GENERATOR OF p

Chooses private key (a)

NOTE: a MUST BE SMALLER THAN p

Chooses private key (b)

NOTE: b MUST BE SMALLER THAN p

Calculates public key (a^*)

$$a^* = q^a \bmod p$$

Calculates public key (b^*)

$$b^* = q^b \bmod p$$

Alice and Bob share a^* and b^*

Computes shared secret key (x)

$$x = (b^*)^a \bmod p$$

Computes shared secret key (x)

$$x = (a^*)^b \bmod p$$

Alice and Bob communicate using a symmetric algorithm of their choice and the shared secret key (x), which was never transmitted over the insecure circuit



Advantages

- No prior shared secret required.
- Enables secure communication over public networks.
- Forms the foundation of modern secure protocols.

Limitation (Important)

- Basic Diffie–Hellman does **not provide authentication**.
- It is vulnerable to a Man-in-the-Middle (MITM) attack unless combined with digital signatures or certificates.

thank you